```r
set.seed(1123)
n <- 10
x <- (1:n)/2
beta_true <- c(10, 3, 2)
sigma_true <- 1/2

g <- function(x, beta) beta[1]*x/(beta[2]+x) + beta[3]

y <- g(x, beta_true) + rnorm(n, 0, sigma_true)
y
```

```
##  [1] 3.780940 5.053554 5.803360 5.544516 7.236928 7.097214 7.364752 8.187043
##  [9] 8.185821 7.678154
```

$$g(x,\beta) = \frac{\beta_1 x}{\beta_2 + x} + \beta_3$$

$$\nabla g = \left( \frac{x}{\beta_2 + x},\ -\frac{\beta_1 x}{(\beta_2 + x)^2},\ 1 \right)$$

```r
library(MASS)

n<-10
x<-1:n/2
beta<-c(10,3,2)
sigma<-1/2
eps<-rnorm(n,0,sigma)
y<-beta[1]*x/(beta[2]+x)+beta[3]+eps

g<-function(x,b1,b2,b3){
  return(b1*x/(b2+x)+b3)}

gradient<-function(x,b1,b2,b3){
  g1<-x/(b2+x)
  g2<- -b1*x/(b2+x)^2
  g3<-1
  return(c(g1,g2,g3))}

G.j<-function(x,b1,b2,b3){
  n<-length(x)
  G<-matrix(rep(0,n*3),ncol=3,nrow=n)
  for (i in 1:n){
    G[i,]<-gradient(x[i],b1,b2,b3)}
  return(G)}

beta.j<-function(x,y,b1,b2,b3){
  beta<-matrix(c(b1,b2,b3),ncol=1,nrow=3)
  G<-G.j(x,b1,b2,b3)
  Y<-matrix((y-g(x,b1,b2,b3)),ncol=1,nrow=length(y))
  beta.nowe<-beta+ginv(t(G)%*%G)%*%t(G)%*%Y
  return(beta.nowe)}

wiarogodność<-function(x,y,b1,b2,b3,sigma){
  n<-length(y)
  return(-n/2*(log(2*pi*sigma))-sum((y-g(x,b1,b2,b3))^2)/2/sigma)}
```

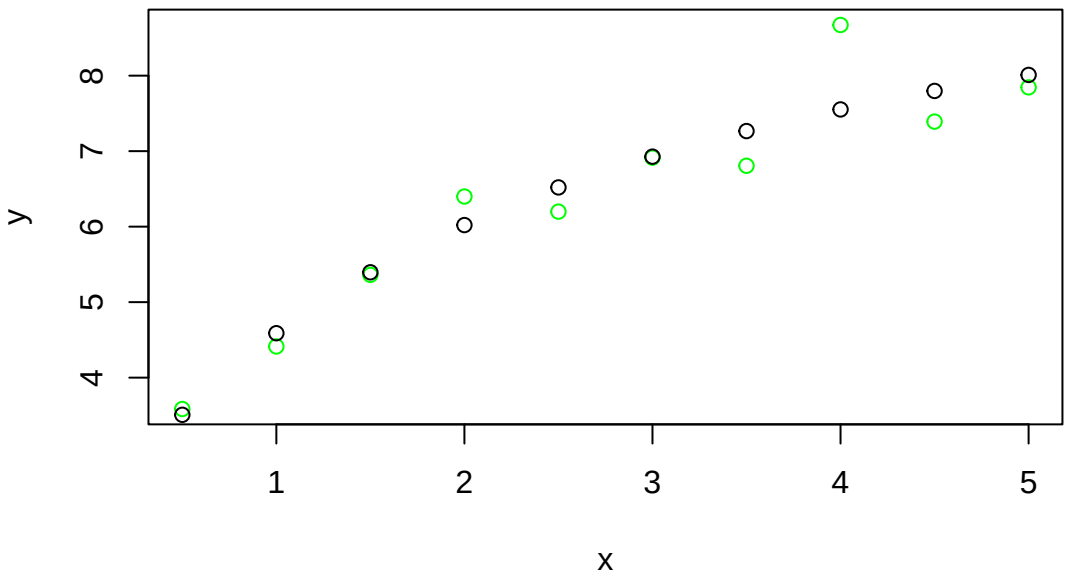
```
Gauss.Newtonn<-function(x,y,b1,b2,b3,M=100,epsilon=0.0001){
  n<-length(y)
  sigma<-sum((y-g(x,b1,b2,b3))^2)/(n-3)
  tabelka<-matrix(c(b1,b2,b3,sigma,
          wiarogodność(x,y,b1,b2,b3,sigma)),ncol=5,nrow=1)
  beta<-matrix(c(b1,b2,b3),ncol=1,nrow=3)
  for (i in 1:M){
    beta.n<-beta.j(x,y,beta[1],beta[2],beta[3])
    sigma<-sum((y-g(x,beta.n[1],beta.n[2],beta.n[3]))^2)/(n-3)
    L<-wiarogodność(x,y,beta.n[1],beta.n[2],beta.n[3],sigma)
    tabelka<-rbind(tabelka,c(beta.n,sigma,L))
    if (sum((beta-beta.n)^2)<=epsilon^2){
      return(tabelka)}
    beta<-beta.n}
  return(tabelka)}
```

```
tab<- Gauss.Newtonn(x,y,b1=8,b2=2,b3=1)
temp2 <- nrow(tab)
tab2 <- tab[c(1,2,3,nrow(tab)),]
temp <- wiarogodność(x, y, 10, 3, 2, 1/4)
rbind(tab2, c(10,3,2,1/4,temp))
```

```
##           [,1]     [,2]     [,3]      [,4]       [,5]
## [1,]  8.000000 2.000000 1.000000 2.0212635 -16.207999
## [2,]  8.905412 2.468109 2.033004 0.2772435  -6.275089
## [3,]  8.971086 2.444911 1.984707 0.2770219  -6.271092
## [4,]  8.972013 2.446759 1.985554 0.2770218  -6.271090
## [5,] 10.000000 3.000000 2.000000 0.2500000  -6.465618
```

Model dobrze przybliża B

```
plot(x,y , col="green")
points(x, g(x, tab[temp2,1], tab[temp2,2], tab[temp2,3]))
```




Model wizualnie dobrze pasuje do danych

```
mx <- tab[temp2,4] * ginv(t(G.j(x, tab[temp2,1], tab[temp2,2], tab[temp2,3])) %*%
  G.j(x, tab[temp2,1], tab[temp2,2], tab[temp2,3]))
mx

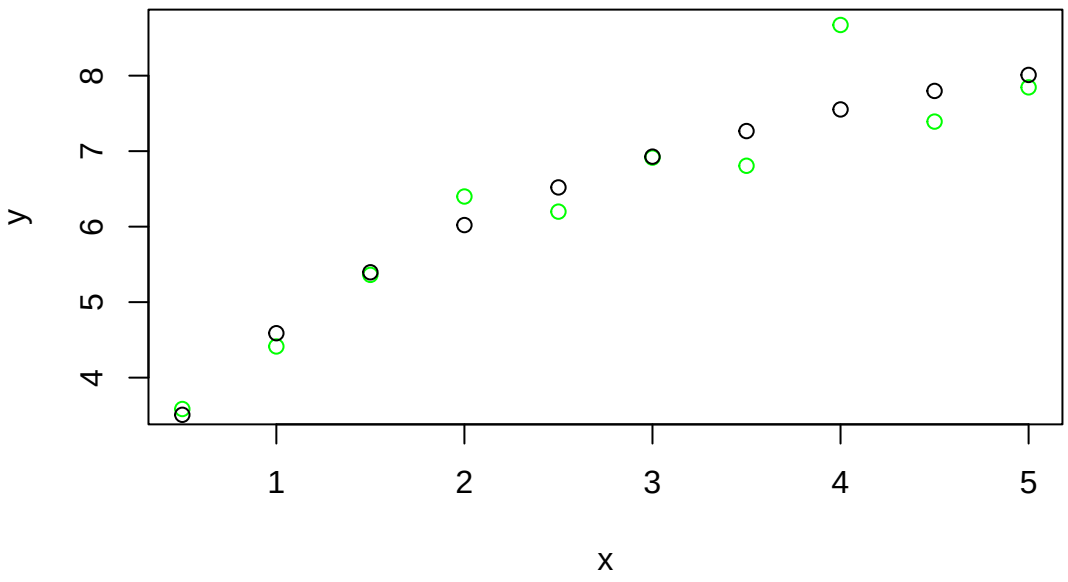
##           [,1]     [,2]      [,3]
## [1,] 2.1301834 2.005905 0.6194671
## [2,] 2.0059055 4.031821 2.3543041
## [3,] 0.6194671 2.354304 1.6714227
```

```
tab<- Gauss.Newtonn(x,y,b1=9.7,b2=2.9,b3=2)
temp2 <- nrow(tab)
tab2 <- tab[c(1,2,3,nrow(tab)),]
temp <- wiarogodność(x, y, 10, 3, 2, 1/4)
rbind(tab2, c(10,3,2,1/4,temp))

##           [,1]     [,2]     [,3]      [,4]      [,5]
## [1,]  9.700000 2.900000 2.000000 0.2861424 -6.433057
## [2,]  8.887000 2.387618 1.982463 0.2771994 -6.274295
## [3,]  8.972823 2.450474 1.988400 0.2770220 -6.271094
## [4,]  8.972012 2.446757 1.985553 0.2770218 -6.271090
## [5,] 10.000000 3.000000 2.000000 0.2500000 -6.465618
```

Model jest dobry, dobrze przybliża prawdziwe wartośći

```
plot(x,y , col="green")
points(x, g(x, tab[temp2,1], tab[temp2,2], tab[temp2,3]))
```

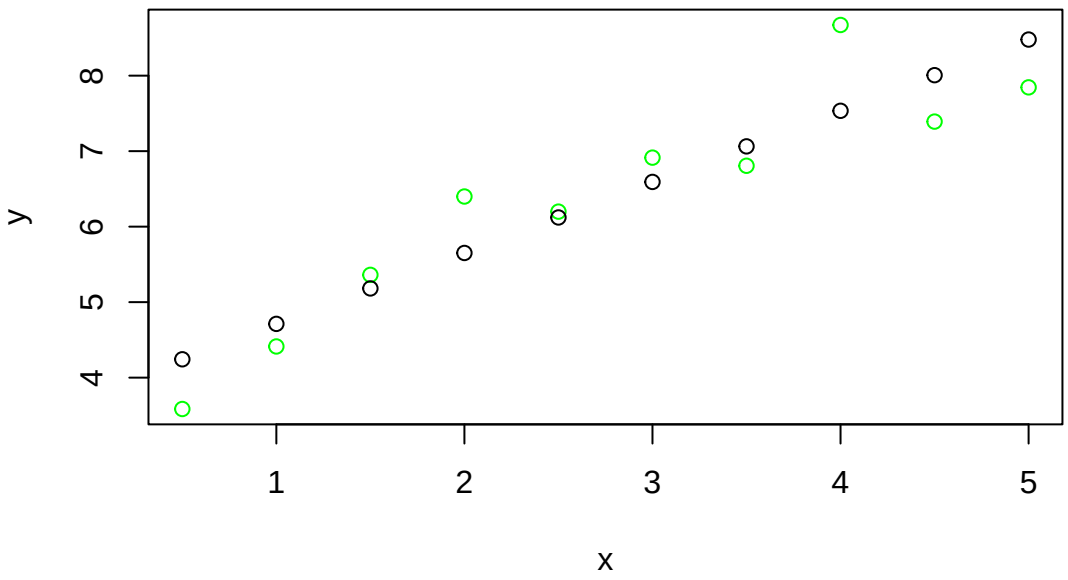



```
tab<- Gauss.Newtonn(x,y,b1=1,b2=1,b3=1)
temp2 <- nrow(tab)
tab2 <- tab[c(1,2,3,nrow(tab)),]
temp <- wiarogodność(x, y, 10, 3, 2, 1/4)
rbind(tab2, c(10,3,2,1/4,temp))

##              [,1]         [,2]     [,3]       [,4]       [,5]
## [1,]     1.000000     1.000000 1.000000 33.7973937 -30.291304
## [2,]     7.413889     9.866864 3.047811  5.5485152 -21.257037
## [3,]    -4.554134   -33.073163 2.267164 21.4805586 -28.025127
## [4,] -1184.624623 -1264.190039 3.774469  0.4801036  -9.020619
## [5,]    10.000000     3.000000 2.000000  0.2500000  -6.465618
```

Model odleciał i wyestymował zupełnie nieprawdziwy wektor B

```
plot(x,y , col="green")
points(x, g(x, tab[temp2,1], tab[temp2,2], tab[temp2,3]))
```


y
4
5
6
7
8
1
2
3
4
5
x